LOT Pager — Firmware Specification

Document: LOT-HW-01 / FW · **Rev:** A (2026-11-13) · **Target:** ESP32-S3, ESP-IDF v5.x (C)

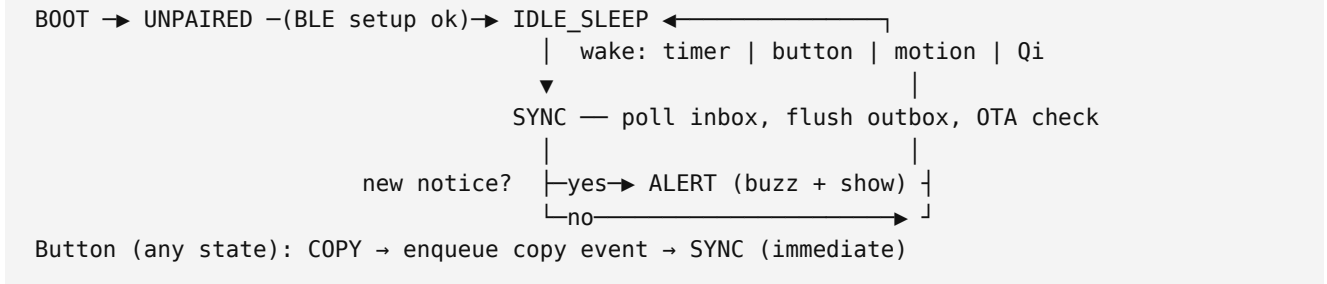
1. Stack

Layer	Choice
SDK / RTOS	ESP-IDF v5.x + FreeRTOS
Network	esp_wifi (station), esp_http_client + mbedTLS, TLS session resumption
Setup	NimBLE (BLE provisioning, `wifi_provisioning` component)
Display	Custom driver for Sharp memory LCD (SPI, 1-bit framebuffer 2 KB) + small bitmap font
Camera	esp32-camera style DVP driver, HM0360 register set; JPEG via software encoder
Sensors	Bosch BSEC2 library (BME688), BHI260AP firmware blob + host driver
Haptics	DRV2605L over I²C, built-in effect library
Identity	ATECC608B via esp-cryptoauthlib
Payloads	CBOR (tinycbor) + **heatshrink** compression
Updates	esp_https_ota, A/B partitions, signed images

Repo layout (to be created in session 2 as `hardware/firmware/`):

```
firmware/
  main/            app_main, state machine
  components/
    lot_api/       HTTPS client for /api/device/*  (doc 05)
    lot_ui/        memory-LCD driver, fonts, screens
    lot_sensors/   BME688 (BSEC2), BHI260AP, VEML6030, MAX17048
    lot_cam/       HM0360 capture + JPEG
    lot_haptic/    DRV2605L patterns
    lot_store/     NVS, outbox queue, compression
  partitions.csv   nvs | otadata | ota_0 | ota_1 | storage
```

2. State machine



3. Events from the button

Gesture	Firmware action	Payload sent
Short press	copy	notice id + text, weather snapshot, battery
Long press 1.5 s	copy + photo	as above + JPEG (QVGA, ~15 KB)

Gesture	Firmware action	Payload sent
Double press	local only	shows BME688 reading

Debounce 30 ms on the snap dome; the button GPIO is an EXT1 deep-sleep wake source.

4. Session compression ("compress the information in each session")

A **device session** = one wake → sync → sleep cycle. Everything that happened since the last sync (button events, sensor samples, battery, errors) goes into one outbox record:

1. Encode as CBOR with short integer keys (no JSON field names on the wire).
2. Sensor series are delta-encoded (temperature in 0.01 °C steps, etc.).
3. Compress with heatshrink (window 8, lookahead 4: ~1 KB RAM).
4. POST once as `application/cbor` with `Content-Encoding: x-heatshrink`.

Typical session: 5 min of BME688 samples + 1 copy event ≈ 420 B JSON → ~110 B on the wire. Fewer bytes = shorter radio-on time = longer battery life. The server decodes and expands to normal Log rows (doc 05).

5. Power budget (80 mAh EVT cell)

Consumer	Assumption	mAh/day
Sleep floor (ESP32-S3 deep sleep + BHI260AP + LCD static)	~35 µA	0.84
Inbox polls	every 10 min 07:00–23:00 (96/day), 0.8 s @ 90 mA with fast reconnect	1.92
BME688	ULP mode, every 5 min	0.10
Button / alerts / camera	10 copies, 10 buzzes, 2 photos	0.40
Total		**≈ 3.3**

→ **~24 days** on 80 mAh (EVT), **~13 days** on 45 mAh (5 mm DVT). Spec: ≥ 7 days. Quiet hours come from the user's timezone on lot-systems.com.

6. Security

- Secure Boot v2 + flash encryption enabled in production builds.
- Device private key generated inside ATECC608B at factory; never leaves the chip.
- Device token (from pairing, doc 05) stored in encrypted NVS; requests signed with the ATECC key (header `X-LOT-Device-Sig`).
- TLS: pin the lot-systems.com CA chain (ISRG Root X1/X2) with a fallback bundle; OTA images signed (ECDSA P-256).

7. Screen UI

- 128 × 128 1-bit. Fonts: one 16 px bitmap font for notices (wraps to 6 lines × 14 chars), one 8 px for status.
- Status row: time · Wi-Fi · battery.
- Screens: `NOTICE`, `COPIED ✓`, `WEATHER`, `CHARGING nn%`, `PAIR CODE: XXXX-XXXX`, `OFFLINE (n queued)`.
- Text only, no images — matches the pager concept.

8. Factory / test firmware

A separate `factory` build (doc 08) runs self-test: I²C scan, LCD pattern, camera frame, BME688 read, haptic pulse, Qi detect, battery gauge, Wi-Fi RSSI, ATECC key gen, then prints a JSON line over the pogo UART for the test jig.